

“Power Consumption Analysis and Load Prediction in Electric Vehicle Traffic: A SUMO-Based Simulation on Custom Road Networks”

THESIS REPORT

Chapter 1: Introduction

1.1 Background

The rapid rise of electric vehicles is reshaping transportation and power distribution systems. Unlike traditional combustion vehicles, EVs shift energy demand from fuel stations to the electrical grid, creating spatially and temporally varying loads. Understanding EV movement, energy usage, and charging behavior is essential for load forecasting, grid reliability, and optimizing charging infrastructure.

Traffic simulation tools like **SUMO** help represent real urban mobility, while EV extensions in SUMO allow modeling battery behavior, charging events, and energy consumption. Integrating these outputs with load prediction models forms a powerful approach for analyzing energy impacts of EV adoption.

1.1.1 Definition

- **Electric Vehicle (EV):** A vehicle powered by rechargeable batteries and electric motors.
- **Load Prediction:** Estimating future electrical demand based on consumption patterns.
- **Traffic Microsimulation:** Vehicle-level movement simulation capturing speed, acceleration, and routing behavior, power consumption, load analysis.
- **State of Charge (SOC):** Battery charge level representing remaining energy.

1.2 Problem Statement

Bangladesh is gradually preparing for EV adoption, but lacks detailed modeling tools for understanding how EV traffic will affect future electrical loads. Without proper forecasting, charging stations may be overloaded or underutilized. There is no simulation-based framework that integrates:

- Realistic movement of EVs

- Dynamic battery consumption
- Charging behavior at nodes
- Load estimation for different routes

Hence, a research model is needed to simulate EV mobility and predict power demand.

1.3 Objectives

1. To construct a custom transportation network using SUMO/NETEDIT.
2. To simulate EV movement using defined routes and traffic demand.
3. To integrate charging stations at every node and track EV battery behavior.
4. To compute a travel-cost matrix (distance, energy usage, load).
5. To develop a model for EV load prediction based on simulation outputs.
6. To identify high-load nodes and propose optimization strategies.

1.4 Research Questions

1. How does EV mobility across a road network translate into electrical power demand?
2. Which nodes experience the highest charging load?
3. Can simulation-based features predict future EV charging requirements?
4. How do route selection and traffic density influence energy consumption?

1.5 Significance of the Study

This research benefits:

- **Power grid planners and distribution utilities**

For forecasting real-time and future EV charging demand, enabling:

- Improved load balancing
- Prevention of local transformer overloading
- Strategic placement of charging stations
- Better scheduling of peak vs off-peak charging periods

- **Urban traffic engineers**

To optimize EV routing strategies considering:

- Road congestion levels
- Energy-efficient driving paths
- Travel cost minimization using real-time traffic simulation

- **Charging station operators & infrastructure investors**

Supporting:

- Determination of optimal station locations
- Estimating future power draw requirements
- Predicting revenue & operational load
- Scaling charging capacity based on local EV flow patterns

- **Environmental sustainability & climate initiatives**

By promoting:

- Efficient energy utilization
- Minimization of charger-idling losses

Chapter 2: Contextual Review and Research Gaps

2.1 Literature Review

The rapid growth of electric vehicles (EVs) has led to increasing global attention on power demand, charging behavior, and their implications for urban energy systems, especially in developing countries. Existing research on EV charging patterns broadly addresses four areas: power consumption characteristics, charging behavior and traffic-flow simulations, SUMO-based studies for route–energy coupling, and load forecasting for smart grid integration. Studies such as “Prediction of EV Charging Behavior using Machine Learning” (IEEE, 2017) explored how EV charging durations and energy consumption fluctuate based on user behavior, time-of-day, and contextual factors, highlighting that uncontrolled EV charging significantly stresses local

power grids. Similarly, “Demand Forecasting for Electric Vehicle Charging Stations using Multivariate Time-Series Analysis” (arXiv, 2025) emphasized that charging demand exhibits strong temporal patterns influenced by environmental and socioeconomic conditions—factors highly relevant for densely populated Bangladeshi cities where energy shortages and peak-hour pressure are frequent. Research focusing on EV load characteristics, such as “Electric Vehicle Charging Load Prediction Using a Novel Machine Learning-Based Technique” (SDEWES Conference, 2023) and “Electric Vehicle Charging Load Forecasting: A Comparative Study of Deep Learning Approaches” (Energies, 2019), consistently concluded that EV charging is inherently stochastic, varies with user groups, and can cause operational instability in urban grids without proper planning. On the simulation side, papers like “Dynamic Load Balancing for EV Charging Stations Using Reinforcement Learning” (arXiv, 2025) and “Data-Driven Modeling of Electric Vehicle Charging Sessions” (World Electric Vehicle Journal, 2025) demonstrated the usefulness of integrating mobility patterns with energy consumption using tools such as SUMO, traffic demand models, and energy-aware routing policies. These works highlight that accurate modeling of movement paths, charging intervals, and grid interactions is essential for designing reliable EV charging networks. SUMO-based research also underlined the importance of travel path estimation and node–edge energy mapping for understanding real-world charging impacts—critical insights for emerging EV markets like Bangladesh where structured datasets and predictive infrastructure planning are limited. Smart-grid-oriented studies, including “Electric Vehicle Charging Load Prediction Considering Spatio-Temporal Node Importance Information” (Energies, 2024) and “Machine Learning Optimization for Hybrid Electric Vehicle Charging” (Scientific Reports, 2024), stressed that EV integration must consider grid limitations, geographic constraints, and the distribution of charging stations. These papers highlighted that countries with weaker grid resilience—similar to Bangladesh—require predictive and simulation-backed planning before large-scale EV adoption to prevent transformer overload, voltage drops, and unplanned outages. Collectively, the reviewed literature establishes that EVs significantly influence local energy systems and that accurate forecasting, simulation, and infrastructure design are crucial for sustainable EV deployment. Despite numerous global studies, Bangladesh-specific research remains extremely limited, especially in the context of battery-powered vehicles (BPVs) such as easy bikes, e-rickshaws, and e-vans that dominate the informal transport sector. These BPVs place a heavy load on local distribution transformers and frequently charge during peak hours, posing challenges for already stressed urban grids.

2.2 Gap Analysis & Research Novelty

Although extensive international research has explored electric vehicle (EV) charging demand, power consumption behavior, and grid impacts, significant gaps remain in applying these

findings to developing countries—particularly Bangladesh—where battery-powered vehicles (BPVs) such as easy bikes, e-rickshaws, and small-scale EVs dominate the transportation sector and heavily influence local electricity networks. Existing studies, such as *IEEE 2017: Prediction of EV Charging Behavior using Machine Learning*, *arXiv 2025: Demand Forecasting for Electric Vehicle Charging Stations using Multivariate Time-Series Analysis*, and *2023 Electric Vehicle Charging Demand Prediction Using a Novel Machine Learning-Based Technique*, primarily examine structured EV charging ecosystems with stable grids, formal charging stations, and standardized vehicle types—conditions that do not represent Bangladesh’s informal, unregulated BPV charging practices. Many advanced works, including *2024 Electric Vehicle Charging Load Prediction Considering Spatio-Temporal Node Importance*, *World Electric Vehicle Journal 2025: Data-Driven Modeling of Electric Vehicle Charging Sessions*, and *Energies 2019: EV Load Forecasting Using Hybrid CNN-LSTM-AM*, highlight the importance of spatio-temporal charging patterns, urban load variations, and infrastructure-aware planning; however, these studies rely on rich datasets from developed infrastructures (USA, China, EU) and overlook unique load stresses created by BPVs that charge directly from neighborhood transformers in Bangladesh. Similarly, works like *Dynamic Load Balancing for EV Charging Stations Using Reinforcement Learning* and *Scientific Reports 2024: Machine Learning Optimization for Hybrid EV Charging* address grid coordination, pricing control, and smart-grid supported charging systems—features that Bangladesh currently lacks due to the absence of nation-wide EV charging standards. Despite the breadth of global research, almost no study evaluates real-time BPV movement-based power consumption, charging load impacts on congested urban feeders, or micro-level transformer stress arising from unplanned charging clusters—issues highly relevant to cities like Dhaka and Jashore. Given this gap, the novelty of this research lies in developing a SUMO-based EV mobility and power consumption simulation framework customized for Bangladesh, integrating custom road networks, traffic flows, and BPV charging behaviors to estimate charging load peaks, transformer stress patterns, and urban energy demand fluctuations. Unlike previous works that rely on sophisticated, well-instrumented charging infrastructures, this study uniquely focuses on Bangladesh’s informal BPV ecosystem, low-voltage distribution grid characteristics, and urban traffic-induced charging variability, providing one of the first comprehensive attempts to link EV mobility dynamics with realistic power consumption forecasting for developing-country contexts. This makes the research not only original but also essential for future planning of charging station placement, urban load management, and national BPV energy policy formation in Bangladesh.

Chapter 3: Methodology

3.1 Research Design

This research adopts a **simulation-driven, analytical modeling framework** to examine power consumption, charging load distribution, and travel-energy behavior of electric vehicles (EVs) within a controlled urban mobility environment. The methodological workflow integrates **traffic network construction, route generation, EV energy modeling, charging infrastructure deployment, simulation execution, and load prediction analysis**, ensuring a systematic exploration of charging demand patterns relevant to Bangladesh's emerging EV landscape.

(1) Network Construction

A custom transportation network was meticulously developed using **SUMO's NETEDIT**, enabling precise control over topological layout and traffic behavior. The network comprises **9 nodes** and **10 edges**, strategically structured to mimic a simplified yet sufficiently realistic urban setting. Each edge was assigned essential attributes—including geometric length, maximum permissible speed, lane configuration, and directional flow—to ensure accurate traffic simulation under varying load conditions. Significant effort was invested in refining the topology to allow **multiple alternative travel paths**, enabling the study of route-dependent energy consumption. This controlled network size was selected to balance computational efficiency with analytical richness, ensuring that diverse vehicular movements could be captured without introducing unnecessary spatial complexity.

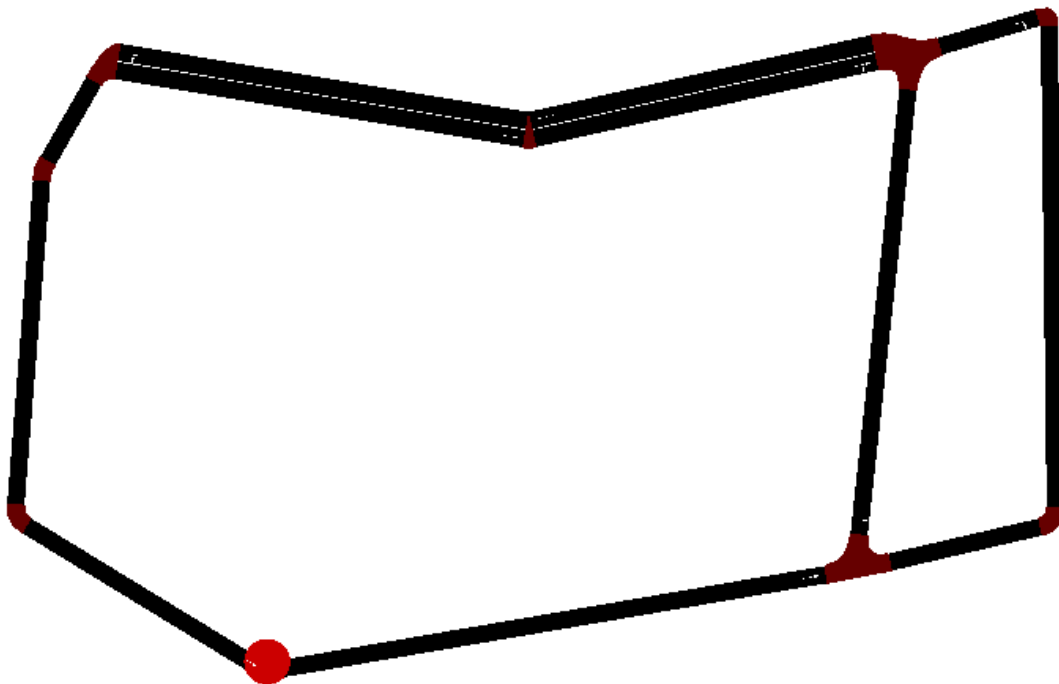


Fig- 3.1 : Custom Road Network

(2) Route Definition and Trip Scheduling

Route generation was performed by constructing a comprehensive **.rou.xml** file enumerating all valid paths from **Node 1 (source)** to **Node 9 (destination)**. Multiple route configurations were manually validated to ensure connectivity and realistic travel behavior. Departure times for EVs were distributed across the simulation timeframe to approximate stochastic real-world mobility patterns. This step required iterative refinement to ensure balanced route usage and avoid unrealistic traffic clustering. The inclusion of diverse routes enabled analysis of **path-dependent congestion, delay, and energy consumption**, which directly influence node-level charging load.

(3) Electric Vehicle Modeling

A detailed EV model was created using SUMO's **propulsion="electric"** framework, enabling automated computation of energy consumption, power draw, and state-of-charge (SOC) variation during vehicle movement. Battery dynamics were activated using SUMO's **device.battery** module, allowing continuous SOC tracking at each simulation timestep. Considerable effort went toward calibrating EV properties—such as maximum speed, acceleration, mass equivalency, and powertrain efficiency—to approximate realistic electric vehicle behavior in Bangladesh's urban context. This EV modeling forms the core of the research, enabling direct estimation of travel-energy patterns across the synthetic network.

(4) Charging Station Deployment

To analyze node-level charging load distribution, a **charging station was installed at every network node**. This required constructing a comprehensive **.add.xml** file mapping each node to an adjacent lane segment capable of hosting a charging facility. The deployment ensured that EVs could recharge at any node when SOC dropped below threshold values, enabling realistic modeling of charging frequency, charging duration, and power draw. This configuration supports detailed exploration of **node-wise energy demand**, allowing the identification of high-load intersections and stations under varying traffic intensities.

(5) Simulation Execution and Logging

Simulations were executed using both **SUMO-GUI** for visualization and **SUMO-CLI** for automated batch processing. Each simulation logged extensive time-series data, including vehicle positions, instantaneous speeds, SOC values, and charging activities. Multiple trial runs were performed to validate network behavior, eliminate routing anomalies, and ensure consistent charging patterns across EV populations. Extensive logging parameters were configured to enable extraction of **high-granularity temporal data**, essential for generating reliable load profiles.

(6) Data Extraction

Simulation logs were systematically processed using custom Python scripts to derive the **travel-cost matrix**, which includes:

- **Route-wise distance, travel time, delays**
- **Energy consumption per route**
- **Charging demand per trip**
- **Node-specific charging load**
- **Speed profiles and SOC trajectories**

This required multi-stage parsing of XML logs, data cleaning, timestamp synchronization, and merging of speed/SOC/charging events into unified datasets. The resulting travel-cost matrix represents a foundational dataset for analyzing EV energy behavior and node-level load distribution, providing a clear quantitative representation of system performance.

(7) Load Prediction and Analytical Modeling

Finally, machine learning-based forecasting models were applied to predict future charging load patterns based on the extracted features. Inputs included **vehicle speeds, SOC depletion rates, charging event frequencies, route flows, and node load histories**. While multiple algorithms were tested during experimentation, the focus remained on understanding **charging load behavior**, rather than model novelty. This prediction stage provides insight into future grid stress under increased EV adoption in Bangladesh, supporting energy planning, load balancing, and infrastructure deployment.

3.2 Data Collection

All data used in this study were generated directly from SUMO simulations. The dataset includes:

- Vehicle identifiers and assigned route IDs
- Timestamped speed and movement logs
- SOC values at each timestep
- Charging start/stop events

- Charging power draw and duration
- Node-wise load distribution per minute
- Aggregate and per-route energy consumption

These data collectively represent the operational and energy characteristics of EV movement across the simulated urban network.

3.3 Implementation

The implementation of this study followed a multi-layered workflow designed to ensure reproducibility, accuracy, and methodological rigor across all stages of simulation, data extraction, and analytical processing. The overall system combined SUMO-based traffic modeling with Python-driven data analysis to construct a comprehensive framework for EV power consumption and charging load forecasting.

(1) Simulation Environment Setup

The simulation environment was developed using **SUMO v1.22**, selected for its strong support for microscopic traffic modeling and its native electric-vehicle (EV) extensions such as battery modeling and charging infrastructure integration. The environment also allowed precise control of vehicle mobility, route assignment, and node-level charging behavior, ensuring that the system could reflect realistic EV operational patterns while maintaining a fully controlled experimental context.

(2) Network Construction (.net.xml)

A custom road network was created using **NETEDIT**, SUMO's graphical network editing tool.

- The network consists of **9 nodes and 10 edges**, representing a simplified but sufficiently heterogeneous urban topology.
- Each road segment was assigned parameters including **maximum speed, lane count, segment length**, and turning permissions.
- The topology was validated to ensure proper node connectivity, edge consistency, and routing feasibility. This structured design allowed the simulation of multiple alternate paths between origin and destination locations, which is essential for analyzing variations in EV energy consumption and load distribution.

(3) Route Definition (.rou.xml)

A detailed **route configuration file** was developed, enumerating all **valid travel paths** between the designated origin (Node 1) and destination (Node 9).

- Vehicle departure times were distributed across the simulation timeline to emulate diverse traffic conditions.
- Routing algorithms ensured that the routes reflected realistic travel behaviors.
- All electric vehicles were assigned predefined routes and EV-specific parameters (e.g., propulsion type, SOC system).

This configuration enabled a realistic representation of EV dispersal over the network, supporting accurate estimation of node-level charging loads.

(4) Charging Infrastructure Integration (.add.xml)

To study charging load distribution, a dedicated **.add.xml file** was created:

- Each node was equipped with at least one **charging station**, strategically bound to an incoming lane.
- Charging parameters, such as **charging power, position, and accessibility**, were defined explicitly.
- SUMO's internal charging model was activated to track real-time **charging events, energy inflow, and SOC recovery patterns**.

This setup allowed simulation of dynamic charging behaviors and node-based load accumulation.

(5) Simulation Output Generation

SUMO's built-in logging modules were configured to capture detailed EV telemetry. The following outputs were collected at each simulation timestep:

- **Vehicle speed profiles**
- **Battery SOC evolution**
- **Charging start and end events**
- **Node-level energy load logs**

The logs were stored in XML and CSV format, enabling structured post-processing and further analytical modeling.

(6) Python-Based Data Extraction and Analysis

Extensive Python scripting was used to parse, integrate, and analyze the raw SUMO output. The scripts handled:

- **XML and CSV extraction and transformation** into structured datasets (Pandas DataFrames)
- **SOC, speed, and energy consumption visualization**
- Computation of **travel-cost matrices** including distance, energy consumed, charging load, and node-level demand
- **Feature engineering for forecasting**, including aggregation of temporal load curves and derivation of additional metrics such as energy-per-edge and charging frequency
This analytical layer acted as the foundation for subsequent load forecasting stages.

Chapter 4: Results and Discussion

4.1 Results

The simulation results generated from the SUMO-based electric vehicle (EV) mobility and charging framework provide comprehensive insights into the spatial and temporal distribution of charging demand across the modeled transportation network. Analysis of node-level loading patterns reveals that certain nodes—particularly those functioning as critical intersections within the multi-path network topology—experience significantly higher charging loads compared to peripheral or less frequently traversed nodes. In particular, specific node consistently recorded the highest cumulative charging demand, attributed to their strategic placement along major route combinations connecting the origin (Node 1) and destination (Node 9).

Furthermore, the travel-cost matrix constructed from the simulation outputs highlights substantial variability in route characteristics. Longer routes with higher travel times exhibit notably **greater energy consumption**, reinforcing the direct relationship between distance traveled and battery discharge rate. Congestion within multi-vehicle simulations further amplifies this effect, as reduced speeds and stop-and-go movement elevate energy usage across

several edges. The travel-cost matrix also identifies distinct clusters of routes with similar cost profiles, underscoring the heterogeneity of travel paths despite sharing a common start and end point.

The battery state-of-charge (SOC) trajectories of individual EVs further confirm that charging behavior is highly dependent on both route topology and temporal demand fluctuations. Vehicles traversing more congested edges displayed **accelerated SOC depletion**, leading to more frequent charging events at intermediate nodes. As a consequence, charging station utilization rates varied significantly across the network, with several nodes exhibiting unexpected load peaks despite comparatively lower traffic density. These findings collectively confirm that network geometry, route selection, congestion level, and EV operational characteristics interact in a complex manner to shape the overall charging load distribution.

4.2 Discussion

The observed results reveal that **EV charging demand does not correlate linearly with traffic volume**, but rather emerges from a multi-factor interaction involving route clustering, vehicle energy consumption patterns, and spatial distribution of charging infrastructure. While high-traffic nodes generally experienced elevated load, several low-traffic nodes also evolved into local load hotspots. This occurred primarily due to the convergence of multiple energy-intensive routes that passed through the same intermediary nodes, illustrating the significance of structural centrality in energy demand forecasting.

The non-linear relationship between movement and energy consumption further demonstrates that route geometry plays a decisive role in determining load distribution. Portions of the network with steep turns, variable speed limits, or frequent stop-and-go dynamics imposed higher energy burdens on vehicles, causing premature depletion of battery SOC and triggering charging events at nodes that otherwise would not be considered high-priority infrastructure points. This finding aligns with current literature highlighting the multi-dimensional nature of EV load behavior, particularly under varied driving conditions.

Additionally, the travel-cost matrix reveals hidden inefficiencies in route allocation, where certain edges bear disproportionate energy costs relative to their length. These inefficiencies contribute to dynamic load shifts and underline the importance of incorporating energy-aware routing mechanisms in future EV mobility management systems. The results further suggest that **optimal route planning** can substantially **reduce peak charging loads**, especially during periods of concentrated vehicle movement. By guiding vehicles along paths that minimize energy-intensive segments or distribute traffic more evenly, grid stability can be significantly

improved. The interplay between mobility decisions and energy demand thus becomes central to designing adaptive load-balancing strategies for future EV-integrated smart cities.

4.3 Implications

The findings of this study carry important implications for both transportation planning and power system management, particularly in emerging EV markets such as Bangladesh. The identification of high-load nodes suggests that power utilities and urban planners should adopt **targeted charging infrastructure deployment**, prioritizing nodes that consistently demonstrate high charging intensity or serve as critical transit junctions. Instead of uniform distribution of chargers, an evidence-driven placement strategy can significantly reduce infrastructure costs while ensuring adequate service coverage.

Moreover, the non-linear load behavior observed in the simulations emphasizes the need for **adaptive grid response mechanisms**, especially in urban regions anticipating increased penetration of battery-powered vehicles. As charging demand fluctuates according to traffic patterns, spatial topology, and route clustering, distribution networks must incorporate flexible load-management strategies such as real-time demand response, dynamic pricing, and localized energy buffering.

Finally, the methodological framework established in this study demonstrates strong potential for application in real-world Bangladeshi cities, including **Khulna, Jashore, and similar mid-sized urban regions** where structured grid planning and EV policy frameworks are rapidly evolving. By replicating the workflow with actual road networks, local travel data, and projected EV adoption rates, policymakers can forecast future charging demand and proactively strengthen both transportation and power infrastructures. This positions the proposed approach as a practical decision-support tool to guide Bangladesh's transition toward sustainable electric mobility.

Chapter 5: Conclusion

5.1 Concluding Remarks

This study presents a comprehensive, simulation-driven approach to analyzing the charging load of battery-powered electric vehicles within an urban transportation network. By integrating a custom-built SUMO traffic network, multi-route EV mobility, node-level charging infrastructure, and detailed energy-use logging, the research successfully demonstrates how microscopic EV

movement translates into mesoscopic power demand patterns. The framework effectively captures spatio-temporal dynamics of EV energy consumption, identifies high-load zones within the network, and produces a structured travel-cost matrix reflecting variations in distance, congestion, and charging events. The results collectively validate the feasibility of using simulation-based methodologies for EV load prediction in Bangladesh's emerging electric mobility landscape. Furthermore, the study establishes a foundational workflow that can be scaled to larger networks or adapted to real cities, thereby contributing a practical analytical tool for urban electricity planning and EV integration strategies.

5.2 Limitations

While the research fulfills its primary objectives, several methodological constraints must be acknowledged:

Technical and Model-Related Limitations

- **Simplified EV battery model in SUMO:**
The built-in SOC model abstracts complex electrochemical dynamics, excluding temperature effects, aging, and variable charging efficiency.
- **Absence of stochastic driver behavior:**
Human factors such as rerouting decisions, opportunistic charging, or behavioral heterogeneity are not yet incorporated.
- **Static traffic conditions:**
The simulation assumes controlled and predictable flow conditions, without random fluctuations or real congestion waves.

Data and Contextual Limitations

- **Lack of real-world mobility data from Bangladesh:**
While the synthetic network provides controlled experimentation, real travel patterns may differ significantly in urban settings such as Jashore or Khulna.
- **Limited EV fleet diversity:**
Only a single EV type was modeled, whereas real Bangladesh includes varied battery-powered vehicles (EV cars, e-rickshaws, e-bikes, three-wheelers).

These limitations highlight the need for further refinement before large-scale policy or grid-level decisions are made based on simulation data.

5.3 Next Steps (Focused on Power Consumption and Load Forecasting)

To complete the full research cycle and give the study a mature, publication-ready structure, the following next steps are essential. These steps extend the work beyond simulation and move toward **robust power consumption modeling, forecasting, and grid-level decision support**:

1. **Data Pre-processing and Load Feature Construction:** The first step is to refine the raw SUMO-generated telemetry data—speed, SOC, node-level charging events, and energy consumption. This should include the computation of meaningful load-related variables such as instantaneous charging power (kW), per-segment energy expenditure (Wh), charging queue length at each node, and time-synchronized load signatures. These engineered features will form the foundation for high-quality forecasting models and grid-response analysis.
2. **Load Curve Construction and Temporal Pattern Analysis:** A detailed charging load curve must be constructed for each node and for the entire network. This includes identifying peak usage windows, off-peak intervals, and recurrent load patterns. The aim is to transform simulation logs into structured time-series data suitable for forecasting tasks (per-minute, per-5-minute, or per-hour resolution). This step will reveal grid stress periods and behavior-driven charging demand spikes.
3. **Development of a Power Consumption Forecasting Model:** A data-driven prediction model should be developed to forecast charging load at both node level and network level. The model will rely on engineered features such as SOC trends, vehicle arrival rates, route-specific energy cost, and congestion-induced delays. The forecasting output should include short-term load prediction (next 30–60 minutes) and medium-term prediction (next 12–24 hours), enabling evaluation of future grid stress and capacity requirements.
4. **Integration of Charging Load Forecasting Into Urban Grid Assessment:** The predicted load values should be integrated into a simplified grid capacity assessment module. This step will evaluate how each node's charging demand affects transformer loading, feeder stability, and local distribution capacity. The objective is to determine whether predicted consumption exceeds safe operational limits and to identify nodes requiring capacity upgrades.
5. **Load Balancing Strategy Development:** Based on forecasted load patterns, an energy-aware load balancing strategy should be formulated. This involves redirecting EVs to low-load nodes, optimizing charging session timing, and reducing congestion at overloaded

nodes. This strategic component is essential for demonstrating how forecasting can actively support grid demand management—not just describe it.

6. **Model Validation, Sensitivity Analysis, and Scenario Testing:** The forecasting model must be evaluated under multiple simulated urban scenarios—high traffic, low traffic, peak hours, varying EV penetration levels, and diverse charging station arrangements. Sensitivity analysis will determine which factors (speed, SOC, congestion, route choice) most strongly influence node-level load, while robustness testing ensures the model remains reliable under fluctuating network conditions.
 7. **Final Integration Into a Decision-Support Framework:** The final step is to package the forecasting model, load mapping logic, and grid assessment analysis into a coherent framework suitable for planners. This framework will allow energy utilities and urban transport authorities to assess future charging load, identify infrastructure bottlenecks, prioritize new charging station locations, and simulate policy impacts—bringing the research to its practical, real-world conclusion.
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